INCH-POUND

MIL-DTL-32045(OS)
22 November 2000

DETAIL SPECIFICATION

CARTRIDGE SET, REDUCED CHARGE PRIMARY

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for the manufacture and acceptance of the Cartridge Set, Reduced Charge Primary. The set consists of 1 reduced charge primary cartridge, P/N 26434-1(2519696-1) and 2 secondary cartridges, P/N 29231-1 (2519696-2) referred to in this document as cartridges.

1.2 Application. The cartridges are used in the Mark 7 ejection seat to provide sequential gas pressure to stroke the ejection gun and force the seat clear of the aircraft.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents specified in Sections 3 and 4 of this specification, whether or not they are listed.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to the Commander, Indian Head Division, Naval Surface Warfare Center, Technical Information Division (Code 840), 101 Strauss Avenue, Indian Head, Maryland 20640-5035, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by sending a letter.

AMSC N/A FSC 1377

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2.2 Government documents.

2.2.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplements thereto, cited in the solicitation (See 6.2).

SPECIFICATIONS

DEPARTMENT OF DEFENSE

MIL-I-8500	Interchangeability and Replaceability of Component Parts for Aerospace Vehicles.
MIL-DTL-21625	Design and Evaluation of Cartridges for Cartridge Actuated Devices

(Unless otherwise indicated, copies of DoD specifications are available from: Standardization Documents Order Desk, Bldg. 4D, 700 Robins Avenue, Philadelphia, PA 19111-5094)

2.2.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

NAVAL AIR SYSTEMS COMMAND DRAWING (Cage Code 30003)

2519696	Cartridge Set, Reduced Charge Primary

PUBLICATIONS

QUALITY ASSURANCE DIRECTORATE (WEAPONS)

AIR 45, Issue No. 3	Proof Schedule,

(Application for copies should be addressed to the Commanding Officer, Naval Air Technical Service Facility, 700 Robins Avenue, Philadelphia, PA 19111-5079.)

2.3 Non-Government publications and drawings. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents, which are DOD adopted, are those listed in the issue of the

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DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in DODISS are the issues of the documents cited in the solicitation (See 6.2).

AMERICAN SOCIETY FOR TESTING MATERIALS (ASTM)

ASTM E 1742	Standard practice for Radiographic Inspection. (DoD adopted)

(Application for copies of ASTMs should be sent to the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959)

AMERICAN SOCIETY FOR QUALITY CONTROL (ASQC)

ASQC Z1.4	Sampling Procedures and tables for Inspection by Attributes. (DoD adopted)

(DoD Activities may obtain copies from: Standardization Documents Order Desk, Bldg. 4D, 700 Robins Avenue, Philadelphia, PA 19111-5094. The private sector and other Government agencies may purchase copies from the American Society for Quality Control, P. O. Box 3005, 611 E Wisconsin Ave., Milwaukee, WI 53201-4606.)

MARTIN BAKER AIRCRAFT COMPANY, LTD. DRAWING (Code Ident U1604)

E/TE – 315	Vented Pressure Vesse l

(Application for copies should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, Attention: Code 9410P, Indian Head, MD 20640-5035)

2.4 Order of Precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supercedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (See 6.2) a sample shall be subjected to first article inspection as specified in sections 4.3 (See 6.3). A first article inspection shall be required if:

a. Two or more years has pasted since the cartridge set or similar items have been manufactured,
b. There has been a major turn over in personnel,
c. There have been major changes to the production facility or
d. There has been a transfer of production to a new facility or site.

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3.2 Design and construction. The design and construction of the cartridge shall be manufactured to drawing 30003-2519696, and shall meet the requirements of this specification with special effort to ensure inherent reliability so that the units will consistently perform as specified. All parts having the same dash number shall be dimensionally and functionally interchangeable per MIL-I-8500.

3.2.1 Item description. The primary cartridge case is made of brass and contains a propellant that is ignited by a percussion primer. The secondary cartridge case is brass with a copper membrane that implodes when heat and pressure are applied. The burning propellants produce the gases necessary to make the ejection gun telescope raising the seat/man for ejection.

3.3 Cartridge production. The cartridge shall be manufactured in a continuous process. Only one lot of primers shall be used in the manufacture of a single lot of cartridges.

3.4 Percussion primers. Primers shall be in accordance with Quality Assurance Directorate (Weapons) Proof Schedule, AIR 45, Issue No. 3, dated 3 May 1971. The primers used shall be from lots manufactured within two years from the assembly date of the cartridge lot.

3.5 Leakage. All cartridges produced under contract shall produce no leaks when leak tested in a dry gas leak detector as specified in 4.4.3.

3.6 Radiographic inspection. All cartridges manufactured under contract shall be free of imperfections in assembly when examined radiographically as specified in 4.2.3.2.

3.7 Ballistic requirements. Cartridges from each first article lot and production lot samples shall meet the requirements listed below when subjected to the tests prescribed in 4.5.

3.7.1 Maximum pressure. The upper limit (UL) for maximum pressure shall be 3,380 psi. The lower limit for maximum pressure shall be 3,000 psi when tested at 70 °F.

3.7.2 Time to maximum pressure. The upper limit (UL) of the time interval from first indication of pressure to maximum pressure shall be 74 milliseconds. The lower limit (LL) for the time to maximum pressure shall be 46 milliseconds.

3.7.3 Time to ½ maximum pressure decrement. The upper limit (UL) for the interval from first indication of pressure to ½ maximum pressure decrement shall be 144 milliseconds. The lower limit (LL) shall be 112 milliseconds.

3.7.4 Misfire. There shall be no misfires.

3.7.5 Hangfire. One or more hangfires of a primary cartridge shall cause rejection of the lot. A hangfire is obtained if the time interval from sear removal to first indication of pressure is greater than 0.050 second.

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3.7.6 Retest. There shall be no retests.

3.7.7 Test failure. If test failure is attributed to an assignable cause, excluding the test cartridges, original test results shall be discarded and that test reconducted.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (See 4.3).

b. Conformance inspection (See 4.4).

4.2 Inspections.

4.2.1 In-process inspections.

4.2.2 Visual inspection. Visually examine all cartridges. Reject those with inadequate or illegible descriptive marking or serial numbers, dent, deep scratches or other defects, which may prevent proper installation of function.

4.2.3 Radiographic inspection. Each cartridge shall be subjected to radiographic inspection.

4.2.3.1 X-ray inspection. Each cartridge shall be X-rayed, in accordance with ASTM E 1742, to verify the presence, condition and proper installation of parts. Any observable imperfections shall be cause for rejection of the unit.

4.2.3.2 Radiograpic procedures. The samples shall be arrayed on trays or boards in consecutive numerical order and each radiograph shall carry a permanent identification of the cartridges displayed thereon. The cartridges shall be positioned to give the most revealing cross-section of each unit. All units shall be identified with serial numbers prior to radiographic inspection. The radiographic identification shall include the complete lot number and the span of serial numbers displayed. Discontinuities in serial numbers shall be identified on the plate. Defective cartridges shall be clearly identified on the plate and shall be removed. Radiographs or copies of the entire production lot shall accompany the ballistic sample to the activity conducting the production lot acceptance test (See 6.2).

4.3 First article inspections. Before entering into quantity production, an acceptable first article sample of 19 cartridge sets shall be prepared using the same procedures and methods proposed for manufacture of the production lot. An acceptable sample is one that has passed all the tests and examinations of 3.3 through 3.6.

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4.3.1 Test methodology. Eighteen of the 19 cartridge sets shall be expended in the tests listed herein and one shall be retained for investigative purposes. First article sample environments, as defined in MIL-DTL-21625, shall consist of table I below. For testing purposes, a set consists of one primary and one secondary cartridge.

Table I. First article environments.

Environment	Quantity
Temperature & humidity Cycling	6
Salt spray	6
Conditioning at –65 °F	3
Conditioning at +200 °F	3
Investigative sample	1

Cartridges shall be fired as described in 4.5 and shall meet the functional requirements of 3.7.

4.4 Quality conformance inspections.

4.4.1 Sampling. Cartridge sets for ballistic testing shall be selected from the production lot in accordance with the requirements of Table I, general inspection level II of ASQC Z1.4 with the exception that the lot size is limited to 500 maximum and the sample size shall be in accordance with table II below. For testing purposes, a set consists of one primary and one secondary cartridge.

Table II. Quality conformance sample size.

Production Lot Size	Ballistic Sample	Investigative Sample	Total Lot Size
0 – 180	15	3	18
181 – 300	20	3	23
301 – 500	25	3	28

4.4.2 Interchangeability. The contractor shall conduct inspections of the production lot to verify total interchangeability of cartridges in accordance with MIL-I-8500 (See 3.2).

4.4.3 Leakage. Each cartridge produced under contract shall be leak test in a dry gas leak tester to determine conformance with 3.5. Units, which exhibit a leak rate in excess

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of 1 X 10^{-5} standard centimeters per second of gas at a pressure differential of 1.0 ± 0.1 atmospheres, shall be rejected.

4.5 Ballistic tests. Cartridges shall be selected in accordance with table II. The sample shall be conditioned to 70 °F for a time period of not less than 6 hours or more than 18 hours. Units shall be fired, within 3 minutes of removal from conditioning, in the Martin Baker test fixture E/TE 315. If not fired within 3 minutes reschedule the test and recondition for 2 hours minimum.

4.5.1 Acceptance criteria. The lot shall be acceptable with respect to functional testing if the cartridges meet the requirements of 3.7. Accept on no defects, reject on one or more.

4.5.2 Misfire. A cartridge that does not fire under the conditions specified in 4.5 shall be considered unacceptable unless the misfire is clearly attributable to faulty test equipment or procedure.

4.5.3 Retest. There shall be no retest.

4.5.4 Test failure. If a test failure is attributed to an assignable cause, excluding the test sample, the original test results shall be discarded and that test reconducted.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order, (See 6.2). When actual packaging of material is to be performed by DOD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department of Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The M397, Cartridge Set, Reduced Charge Primary is designed to generate hot gases to extend the telescoping piston to eject the man/seat from military aircraft. This primary cartridge set is military unique and there is no equivalent commercial application for this design.

6.2 Acquisition requirements. Acquisition documents must specify the following:

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a. Title, number and date of this specification.

b. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (See 2.2.1 and 2.2.2).

c. Whether first article inspection is required and, if so, specify the test activity (See 3.1 and 4.3).

d. Production lot size (See 4.4.1) and test activity.

e. Packaging requirements (See 5.1 and 6.5).

f. Safety requirements (see 6.4).

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, or a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested (see 4.3). The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles.

6.3.1 Bids. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Ammunition and explosive safety.

6.4.1 Contractor's responsibility. The safety requirements of the "Contractor's Safety Manual for Ammunition, Explosives and Related Dangerous Material" (DOD 4145.26M) are applicable when this specification is used for acquisition.

6.4.2 Government activity's responsibility. The safety precaution requirements of "Ammunition and Explosives Ashore", OP5 are applicable when this specification is used as part of a description of work to be accomplished by a government activity.

6.5 Contract packaging, packing and marking. The following requirements should be specified in the contract.

6.5.1 Packaging and packing. In addition to the following, Level A and Level C packaging and packing should be in accordance with 49 DFR 100-199.

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6.5.1.1 Level A. Level A packaging and packing should be used for packaging and packing of all production lots for service use. For an explanation of terms see 6.6. and 6.7.

6.5.1.1.1 Inner container. A maximum of one cartridge set, consisting of one Reduced Charge Primary cartridge and two Secondary cartridges shall be packaged in a sealed metal container which conforms to the requirements of MIL-P-116, Method IA-5, or a sealed metal container, which will provide equal protection.

6.5.1.1.2 Outer container. Twenty inner containers shall be packed in the outer container conforming to PPP-B-621, Class 2, Style 4 or 4½ or a box conforming to PPP-B-601, Overseas Type, Style A or B or equal.

6.5.1.2 Level C. Minimum packaging and packing is required for shipping those cartridges, which are intended for expenditure in production lot acceptance tests. A tight pack, is recommended

6.5.2 Marking.

6.5.2.1 Special marking. Marking of outer containers should be in accordance with CFR 49 171-199.

6.5.2.2 Normal marking. In addition to any special marking required by the contract or order, unit packages, intermediate packages and shipping containers should be marked in accordance with MIL-STD-129.

Inner container marking.

NSN: 1377-00-845-5242 DODIC: M397

Nomenclature: Cartridge Set, Seat Ejection

Part Number: 30003-2519696

Lot Number: (In accordance with MIL-STD-1168)

ICC Explosive Class: C

Outer container marking.

NSN: 1377-00-845-5242 DODIC: M397

Nomenclature: Cartridge Set, Ejection Seat.

Part Number: 30003-2519696

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Quantity: 20

Gross weight and cube:

Lot number: (In accordance with MIL-STD-1168)

Contract or purchase order number:

“ WARNING: EXPLOSIVE POWER DEVICE, CLASS C EXPLOSIVE, HANDLE WITH CARE, KEEP FIRE AWAY ”

The above information shall be applied in accordance with the provisions of MIL-STD-129.

6.6 Definitions.

6.6.1 Level A packaging. The degree of preservation and packaging that will afford adequate protection against corrosion, deterioration and physical damage during shipment, handling, indeterminate storage and world wide redistribution.

6.6.2 Level C packaging. The degree of preservation and packaging that will afford adequate protection against corrosion, deterioration and physical damage during shipment form supply source to the first receiving activity for immediate use. This level may conform to contractors commercial practice when such meets the requirements of this level.

6.6.3 Level A packing. The degree of packing that will afford adequate protection during shipment, handling, indeterminate storage and worldwide distribution.

6.6.4 Level C packing. The degree of packing that will afford adequate protection against corrosion, deterioration and physical damage during shipment form supply source to the first receiving activity for immediate use. This level may conform to contractors commercial practice when such meets the requirements of this level.

6.7 Explanation of terms. For the purposes of the specification, NAVSUPINST 4030.28A should be used to clarify terms.

6.8 Subject term (key word) listing.

Aircraft Ejection seat
Emergency egress

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Cartridge
Gas Generator

Custodian:
Navy ? OS

Preparing Activity:
Navy – OS
(Project No. 1377-0108)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-DTL-32045	2. DOCUMENT DATE (YYMMDD) 22 November 2000

3. DOCUMENT TITLE
CARTRIDGE SET, REDUCED CHARGE PRIMARY

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (*Last, First, Middle Initial*)	b. ORGANIZATION	
C. ADDRESS (*Include Zip Code*)	d. TELEPHONE (*Include Area Code*) (1) Commercial (2) DSN (*If applicable*)	7. DATE SUBMITTED (*YYMMDD*)

8. PREPARING ACTIVITY

a. NAME Commander, Indian Head Division Naval Surface Warfare Center	b. TELEPHONE (*Include Area Code*) (1) Commercial: 301-744-1973 (2) DSN: 354-1973
c. ADDRESS (*Include Zip Code*) Engineering Documentation Branch (Code 4230) 101 Strauss Avenue Indian Head, MD 20640-5035	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Standardization Program Office (DLSC-LM) 8725 John J. Kingman Road, Suite 2533 Fort Belvoir, VA 22060-6221 Telephone (703) 767-6888 DSN 427-6888

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